University of Minnesota custom Azure course

Lab 3 Solution: Immutable Backup Vault with MUA

Instructor solution and facilitator checks

Session 3: Data Protection in Both Directions

Session: Day 2, Session 3, Block 4

Audience: Instructor during delivery; attendees after the course

Includes: Expected output for every checkpoint, refusal messages, the as-code solution, debrief model answers, failure modes

Keep this document out of the participant folder until the course ends.

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1. How to use this solution

Companion to Lab 3. It gives the expected output for each checkpoint, the exact shape of the refusals in Part 5, the as-code path, the debrief model answers, and the failure modes from dry runs. The lab has one long-running dependency (the first vaulted backup); the sequencing notes in section 5 keep the room moving while it runs.

> **Two ways to reach the end state**
>
> **Step by step**: sections 2 to 7 follow the participant guide.
>
> **As code**: `lab-solutions/s03-immutable-backup-vault/deploy.sh` deploys `guard.bicep` into the guard group and `main.bicep` into the lab group, seeds the containers, configures the backup instance, starts the backup, attempts a protected operation, and prints the restore commands. Section 8.

2. Part 1: Resource Guard

Expected `list-protected-operations` output (Backup vault resource type) includes operations such as:

```
Microsoft.DataProtection/backupVaults/backupInstances/delete
Microsoft.DataProtection/backupVaults/write   (changes to soft delete, immutability, and
security settings)
Microsoft.DataProtection/backupVaults/backupResourceGuardProxies/delete
Microsoft.DataProtection/backupVaults/backupInstances/stopProtection/action
... (reduce retention on policies and instances, delete recovery points)
```

The exact list evolves; read it from the command rather than from this page when a question is asked. The point for the room: every operation that could lose a recovery point, or remove a control that prevents losing one, is on the list.

3. Part 2: vault and MUA link

Expected Checkpoint 2:

```
Immutability    SoftDelete    Days
--------------  ------------  ------
Unlocked        On            14
/subscriptions/.../resourceGroups/rg-lab-07-
guard/providers/Microsoft.DataProtection/resourceGuards/guard-lab03-ajs07
```

Facilitator note: the MUA link requires the caller to have permission on the guard. Attendees hold Contributor on the guard group, which includes read; the link succeeds. If the instructor pre-created

guards in a group the attendees cannot read, the mapping create fails with an authorization error, which is itself a useful moment but is not the lab's design.

4. Part 3: the protected account

The account enables versioning and change feed deliberately: vaulted backup does not need them, but if an attendee adds operational backup later for comparison, those are prerequisites (and Backup enables them on the account anyway). A Backup-owned delete lock appears on the account once it is protected; it is removed when the instance is deleted.

5. Part 4: vaulted backup and sequencing

The two-step initialization is the part attendees most often get wrong by using the operational-tier pattern (`initialize` alone). Expected content of `backup-config.json`:

```
{
  "containers_list": ["research-active", "research-reference"],
  "include_all_containers": false,
  "object_type": "BlobBackupDatasourceParameters",
  "storage_account_name": "stlab03ajs07",
  "storage_account_resource_group_name": "rg-lab-07"
}
```

If `validate-for-backup` fails with an authorization error, the role assignment has not propagated; the lab's advice is to retry every few minutes while continuing with Part 5 (Part 5 does not depend on the backup). If it fails because the account has no containers, Part 3 was skipped.

Keeping the room moving

- Announce at the start of Part 4 that the backup runs for 20 to 40 minutes and that Part 5 happens while it runs.
- Attendees whose backup finishes early do Part 6 immediately; others do it after the debrief. Both groups have the recovery point by the end of the session.
- The job list is the status board: `az dataprotection job list -g $RG --vault-name $VAULT -o table`.

6. Part 5: the refusals

Each protected operation attempted without guard access is rejected. The message names the Resource Guard; typical text:

```
(ResourceGuardUnauthorized) The operation is protected by Resource Guard
'/subscriptions/.../resourceGuards/guard-lab03-ajs07'.
The caller does not have sufficient permission on the Resource Guard to perform this
operation.
```

The message wording varies by release; what is constant is the error code family and the guard reference. Have attendees write the first line.

Restoring guard access

Attendees cannot re-grant themselves a role on a group where they no longer hold User Access Administrator. The guide tells them to ask you. Keep this one-liner ready; it takes seconds per attendee:

```
az role assignment create --role Contributor --assignee <attendee-object-id> --scope
/subscriptions/<sub>/resourceGroups/rg-lab-NN-guard
```

Alternative design if the room is large: give each attendee a second test identity with Contributor on the vault group only, and have them run Part 5 as that identity (`az login` in a second terminal). Then no roles need to be removed or restored.

7. Part 6: restore

Expected job list after the restore is triggered:

```
Op       Status
-------  ----------
Backup   Completed
Restore  InProgress
```

`initialize-for-item-recovery` with `--source-datastore VaultStore`, `--recovery-point-id`, `--target-resource-id` (the alternate account), and `--container-list` is the vaulted-tier restore request; the operational tier uses `--point-in-time` instead of a recovery point and restores to the source account only. The vault identity must hold Storage Account Backup Contributor on the alternate account, which the guide grants before the restore.

Restore completion for five small blobs is a few minutes. Listing the alternate account's container needs a data role for the attendee (Storage Blob Data Reader) on that account; the guide notes it.

8. The as-code solution

```
export LAB_RG=rg-lab-07 LAB_GUARD_RG=rg-lab-07-guard LAB_SUFFIX=ajs07
cd lab-solutions/s03-immutable-backup-vault
./deploy.sh
```

File	What it deploys	Notes
`guard.bicep`	Resource Guard in the guard group	No exclusions, so every supported operation is protected

File	What it deploys	Notes
`main.bicep`	Storage account (versioning, change feed, two containers), Backup vault (immutability Unlocked, soft delete On 14 days, system identity, daily vaulted blob policy P30D), MUA proxy, Storage Account Backup Contributor for the vault identity	The proxy resource `backupResourceGuardProxies/DppResourceGuardProxy` is the Bicep form of the CLI `resource-guard-mapping create`
`deploy.sh`	Seeds blobs, runs the two-step backup instance initialization, creates the instance, starts an on-demand backup, attempts a protected operation, prints the restore sequence	The restore commands are printed rather than run because the first backup must complete first

The backup instance is not expressed in Bicep on purpose: the CLI's `initialize-backupconfig` and `initialize` produce the request bodies that the documented path expects, and the resulting JSON can be reviewed by attendees. (A `backupInstances` child resource is possible in Bicep for teams that want everything declarative; its body mirrors `backup-instance.json`.)

9. Debrief: model answers

Question	Representative answer	What to listen for
Which control would have stopped the attack you worry about most?	Compromised backup admin deleting backups: immutable vault. Insider shortening retention to let backups expire: immutability plus MUA. Deleting the vault outright: MUA and soft delete. Encrypting the source: none of these prevent it, which is why recovery points exist.	A specific attack mapped to a specific control

Question	Representative answer	What to listen for
Who should own the Resource Guard?	Security or a governance function, in a subscription (or tenant) where backup and platform administrators hold no roles. Their access to the vaults should be read-only, so that no single team can both operate and unprotect.	Separation as an organizational fact, not a setting
What does 'immutable' mean now?	Our backups cannot be deleted, shortened, or switched off by any single administrator, including a compromised one, until their retention expires.	Precision and the word 'single'
When is operational backup still right?	Fast, point-in-time recovery of active blob data inside the source account, where the threat is accidental overwrite or a bad batch rather than an attacker with account rights; pair it with vaulted backup for the offsite copy. Not for hierarchical namespace accounts, which use vaulted backup only.	Understanding the two tiers as complements
What does the slow first backup imply?	Sequence protection by criticality and size, start it early, and schedule restore tests after the first full backup completes; measure actual restore time against RTO rather than assuming.	Operational realism

10. Dry-run failure modes

Symptom	Cause	Fix
`backup-vault create` rejects `--storage-setting`	Quoting	Use the exact string in the guide, including the brackets, inside double quotes
`resource-guard-mapping create` fails with authorization error	Attendee lacks read on the guard	Confirm Contributor on the guard group; wait for propagation
`validate-for-backup` fails repeatedly	Role propagation or missing containers	Retry every five minutes; confirm two containers exist
Part 5 operations succeed	Attendee still holds a role on the guard group (often through a group membership)	`az role assignment list --assignee $ME --scope $RGG_ID --include-groups`; remove, retest
Attendee cannot restore their own guard access	Expected	Instructor re-grants Contributor on the guard group

Symptom	Cause	Fix
`restore trigger` fails on the target	Vault identity lacks the role on the alternate account, or the recovery point is from a job that has not completed	Grant Storage Account Backup Contributor on the alternate account; wait for the Backup job to complete
Cleanup: `backup-instance delete` refused	Guard access not restored, or the instance is still backing up	Restore access; wait for the job

11. Appendix: the backup instance as Bicep

Teams that want the whole protection path declarative can express the backup instance as a child resource of the vault. The body below mirrors what `initialize` produces; the vault identity must already hold Storage Account Backup Contributor on the account, and the deployment must run after that assignment has propagated (a `dependsOn` on the role assignment plus a short deployment script delay is the usual pattern).

```
resource backupInstance 'Microsoft.DataProtection/backupVaults/backupInstances@2024-04-01'
= {
  parent: backupVault
  name: '${storageAccountName}-vaulted'
  properties: {
    objectType: 'BackupInstance'
    friendlyName: storageAccountName
    dataSourceInfo: {
      datasourceType: 'Microsoft.Storage/storageAccounts/blobServices'
      objectType: 'Datasource'
      resourceID: storageAccountId
      resourceLocation: location
      resourceName: storageAccountName
      resourceType: 'Microsoft.Storage/storageAccounts'
      resourceUri: storageAccountId
    }
    policyInfo: {
      policyId: blobPolicyId
      policyParameters: {
        backupDatasourceParametersList: [
          {
            objectType: 'BlobBackupDatasourceParameters'
            containersList: [ 'research-active', 'research-reference' ]
          }
        ]
      }
    }
  }
}
```

Validate the body against the live schema with `az bicep build` before use; the `policyParameters` block is where vaulted and operational differ, and the Azure Resource Manager template reference for `backupInstances` is the source of truth.

12. Instructor notes on the lab environment

- Each attendee needs two resource groups (`rg-lab-NN` and `rg-lab-NN-guard`) with Contributor and User Access Administrator on both; the setup guide's section 9 loop creates them.
- Register `Microsoft.DataProtection` on the lab subscription before the course; the first `backup-vault create` in a subscription without it fails with a provider error.
- Pre-install the `dataprotection` extension in Cloud Shell images by having attendees run `az extension add --name dataprotection` during Part 0; the extension pulls in the current command set (`initialize-backupconfig` with `--container-list`, then `initialize` with `--backup-configuration`) used in the guide.
- Keep the one-liner that restores guard access ready in a terminal (section 6); it is the only step attendees cannot do for themselves.
- Vault cleanup: an Unlocked vault with soft delete On keeps deleted instances for 14 days; delete the vaults in your end-of-course pass after that window. Attendee resource group deletion fails until then, which is expected, so tell the room during the debrief.
- Cost: Backup vaults bill for protected instances and vault storage; ten attendees' vaults with ten small blobs each are negligible, and the LRS vault store keeps it that way.

Timing observed in dry runs

Step	Typical duration	Note
Resource Guard create	under 1 minute	
Backup vault create	1 to 2 minutes	Identity creation adds a few seconds
Role propagation to the vault identity	2 to 10 minutes	Up to 30 minutes per Microsoft Learn
First vaulted backup, ten 512 KB blobs	15 to 35 minutes	Dominated by job orchestration, not data volume
Restore of one container to the alternate account	3 to 8 minutes	
Protected operation refusal	immediate	Evaluated before the operation starts